MAS · Vector Calculus and Motion — Patterns and Solution Approaches

Based on: 12 past exam questions from 2016–2025 with marking keys + examiner reports. Appears only in the Calculator-assumed paper. Original questions: see `topics/04_Vector_Calculus.pdf`. Review status: has undergone one round of independent review and was revised according to feedback (see `Review_log.md`). On 2026-10-09, per user feedback, 2017A-Q18 and 2019A-Q18, which do not involve vectors, were moved out of this topic (into "Rates of Change and Differential Equations").

Marking-key reminders

- Difference between distance travelled and displacement vector (2025 report focus): $\int_a^b v(t)\,dt$ is the change in displacement vector (has direction), while $\int_a^b |v(t)|\,dt$ is the distance travelled (scalar).
- The constant of integration is a vector (2018 report: many students did not clearly show how to use the constant of integration to derive r(t)): $r(t) = \int v\,dt + c$, then use r(0) to solve for each component of c.
- Notation: vectors must be marked (tilde or bold); velocity vector and speed must be written separately; the keys for 2019A-Q13 and 2022A-Q10 each award 1 mark specifically for notation.
- After eliminating the parameter to obtain a Cartesian equation, you must state the restrictions on the variables (often both x and y must be restricted) (the keys for 2016A-Q16, 2024A-Q18 and 2025A-Q12 each award this mark).

Formula card

Quantity	Expression
Velocity	$v(t) = r'(t)$ (differentiate component by component)
Acceleration	$a(t) = v'(t)$
Speed	$\lvert v(t)\rvert = \sqrt{(\dot{x}^2 + \dot{y}^2 (+ \dot{z}^2))}$
Distance travelled (arc length)	$\int_a^b \lvert v(t)\rvert\,dt$
Change in displacement	$\int_a^b v(t)\,dt = r(b) - r(a)$
Direction of motion	angle with the x-axis $\tan^{-1}(\dot{y}/\dot{x})$, note the quadrant

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Pattern 1: Given r(t), find velocity, acceleration and speed

Approach: differentiate component by component → substitute the time → take the magnitude for speed → draw the vector on the diagram (place its tail at the position at that time, with the arrow pointing in the direction of motion). Constant speed test (2020A-Q12): $r = (10\cos 0.1t, 10\sin 0.1t, 0.2t) \Rightarrow v = (-\sin 0.1t, \cos 0.1t, 0.2) \Rightarrow |v| = \sqrt{(\sin^2 + \cos^2 + 0.04)} = \sqrt{1.04} \approx 1.02$ m/s, independent of t ⇒ constant speed, but the direction of the velocity vector is always changing, so it is not uniform straight-line motion. You must state that $\sin^2 + \cos^2 = 1$ was used (1 mark on its own). Observation deck height = 0.2 × 65 = 13 m.

Past questions: 2016A-Q16(b), 2017A-Q15(a)(d), 2019A-Q13(a)(b), 2020A-Q12, 2023A-Q16(a), 2024A-Q18(b)(d).

Pattern 2: Given a(t) or v(t), integrate to find r(t)

Standard procedure

1. Integrate component by component, adding the constant vector $c = (c_1, c_2)$.
2. Use the initial conditions (r(0), v(0)) to solve for each component.
3. Write the complete r(t).

Projectile motion, two models:

- Gravity + wind (2018A-Q19): $a = (0, -9.8) \Rightarrow v = (70\cos\theta + w, 70\sin\theta - 9.8t) \Rightarrow r = ((70\cos\theta + w)t, 70\sin\theta \cdot t - 4.9t^2)$. Wind affects only the horizontal component.
- Horizontal drag + vertical lift (2021A-Q19): launched horizontally from E(100, 120); $x' = 32e^{-0.05t}$; vertical acceleration $h'' = s - 9.8$ (where s is the lift), and back-solve for $s = 4.8$ from the given r, so $r = (740 - 640e^{-0.05t}, 120 - 2.5t^2)$. Range or landing point: $y(t) = 0$ (or equal to the height of the slope) ⇒ solve for t ⇒ substitute into x(t). Maximum range ⇒ differentiate with respect to θ or use CAS to find the extremum. Landing angle: use the angle between the velocity vector at the instant of landing and the slope or horizontal plane (2021A-Q19(e)).

Past questions: 2017A-Q15(c), 2018A-Q19(a)(c), 2021A-Q19, 2022A-Q10(b).

Pattern 3: Eliminate the parameter to obtain a Cartesian equation ★

Common identities

- $\cos^2 t + \sin^2 t = 1 \Rightarrow$ circle or ellipse (2022A-Q10(c));
- $\cos 2t = 1 - 2\sin^2 t$ or $2\cos^2 t - 1$, directly gives the relationship between x and y ⇒ a segment of a parabola (2016A-Q16: $x = 2y^2 - 4$);
- first use double-angle formulas to reduce both components to the same angle, then use $\sin^2 + \cos^2 = 1$ to eliminate the parameter ⇒ a general curve (2019A-Q13(d): $(1 - y)^2 + (x^2/2 - 1)^2 = 1$, a figure-eight);
- x is linear in t: solve for t directly from x and substitute into y (2018A-Q19(b); the 2018 report singled out this part as poorly answered);
- $(e^t + e^{-t})^2 - (e^t - e^{-t})^2 = 4 \Rightarrow$ hyperbola $x^2 - y^2 = 4$ (2024A-Q18(e));
- exponential parameter: solve $t = \ln(\ldots)$ from x, then substitute into y (2025A-Q12(a)).

Do not omit the final step: state the range of the actual path, deduced from the range of t; the keys state: 2016A-Q16: $-2 \le y \le 2$ (or $-4 \le x \le 4$); 2024A-Q18: $x \ge 2$ and $y \ge 0$ (writing only $x \ge 2$ would include the lower branch and lose the mark); 2025A-Q12: $1 \le x \le e^3$.

Past questions: 2016A-Q16(a), 2018A-Q19(b), 2019A-Q13(d), 2022A-Q10(c), 2024A-Q18(e), 2025A-Q12(a).

Pattern 4: Interpreting distance travelled (arc length) and displacement

Approach

1. Find the start and end times. The time for one circuit = the least common multiple of the periods of the components: 2022A-Q10 is 2π, 2019A-Q13 is 4π (contains t/2), 2017A-Q15 is 6π (contains t/3). These two questions each award 1 mark for correct limits. You can also back-solve for t from the coordinates: point B is (10/3, 8/3) at $t = \ln 3$.
2. Write $\int_a^b \sqrt{(\dot{x}^2 + \dot{y}^2)}\, dt$ (the usual marking points: the start and end t-values or limits, the speed expression, and the calculated result or notation; the allocation varies by year; "Do not evaluate" questions have no mark for calculation).
3. When the question says "Do not evaluate", just write the integral expression containing the trigonometric functions.

Explain the meaning of the integrals (2 marks)

- $\int_0^1 v(t)\, dt$: "the change in displacement (position vector) during the first second".
- $\int_0^{2\pi} |v(t)|\, dt$: "the distance travelled in one complete circuit (2π seconds)" (2022A-Q10).
- $\int_0^3 r'(t)\, dt$: the change in displacement vector, drawn on the diagram as an arrow from r(0) to r(3) (2025A-Q12(c)).

Past questions: 2016A-Q16(c), 2017A-Q15(b)(e), 2019A-Q13(c), 2022A-Q10(a), 2023A-Q16(b), 2024A-Q18(c), 2025A-Q12(b)–(d).

Pattern 5: Closest distance between objects moving in straight lines (3D)

Approach

1. Write the positions of the two objects: $r_K(t) = r_{K0} + t \cdot v_K$, $r_B(t) = r_{B0} + t \cdot v_B$.
2. Relative position $d(t) = r_K(t) - r_B(t)$. The position of P relative to Q = r_P − r_Q; when the question asks for a relative position vector, the order must be correct (2017A-Q14(b): "from the top of the tower" ⇒ r_drone − r_tower, the key awards 1 mark). When only distance is required, the order does not affect the magnitude.
3. Closest distance: set $d(t) \cdot (v_K - v_B) = 0$, or use CAS to find the minimum of $|d(t)|$.
4. "Time interval within 50 m": solve $|d(t)| < 50$ and find the length of the time interval (2017A-Q14(c)).
5. Shortest distance between two paths (2024A-Q16(c): the speed and departure time can be adjusted, so the problem reduces to the shortest distance between two straight lines): use two independent parameters λ, μ, and require the common perpendicular vector to be perpendicular to both direction vectors (2024A-Q16(c)).
6. Angle of descent (angle with the horizontal plane): first find the angle φ between the velocity vector and k; angle of descent = $|90° - \varphi|$ (Ben's v = (−0.5, 1, −1.5), φ ≈ 143.3°, 90° − φ = −53.3°, answer 53°; 2024A-Q16(b)). You can also use $\sin^{-1}(|v_z| / |v|)$ directly.

(The Ferris wheel and carousel questions (2017A-Q18, 2019A-Q18) do not involve vectors and have been moved to the related rates pattern in the "Rates of Change and Differential Equations" notes.)

Past questions: 2017A-Q14, 2024A-Q16.